

Bayes' Rule

The presence of prior probability (belief)



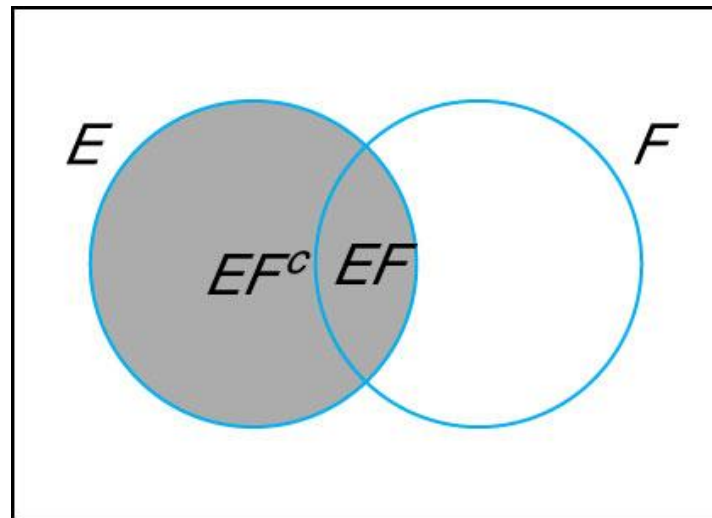


Same picture upside down

Suggested explanation: Based on prior experience, we usually think that the light comes from above. This is our “prior probability”

Bayes' Formula

Discovered by Reverend Thomas Bayes (1702-1761), an English priest interested in philosophical and mathematical problems



Let E and F be events. We may express E as

$$E = EF \cup EF^c$$

For in order for a point to be in E , it must either be in both E and F or be in E but not in F . As EF and EF^c are mutually exclusive, by Axiom 3

$$\begin{aligned} P(E) &= P(EF) + P(EF^c) \\ &= P(E|F)P(F) + P(E|F^c)P(F^c) \\ &= P(E|F)P(F) + P(E|F^c)[1 - P(F)] \end{aligned}$$

Physical meaning

- The probability of the event E is a **weighted average** of the **conditional probability of E given that F has occurred $P(E|F)$** and the **conditional probability of E given that F has not occurred $P(E|F^c)$** , with each conditional probability being given as much weight as the event it is conditioned on has of occurring
- An extremely useful formula, for its use often enables us to determine the probability of an event by first **“conditioning” on whether or not some second event has occurred**, as the examples below show.

Example

Steve answers a multiple choice question either by knowing the answer or guessing. Suppose the probability that he knows the answer is 0.8 and the probability that he guesses is 0.2. Let there be 4 choices in the question. What is the conditional probability that **he knows the answer**, given that **he answers correctly**?

Let E be the event that the Steve knows the answer.
Let F be the event that he answers correctly.

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(EF) = P(F|E)P(E)$$

$P(E) = 0.8$ (There is a probability of 0.8 that he knows the answer)

$P(F|E) = 1$ (He will surely answers the question correctly if he knows the answer)

Hence $P(EF) = 0.8$

F can be divided into two mutually exclusive events: He knows the answer or guesses. By applying Bayes' formula

$$P(F) = P(F|E)P(E) + P(F|E^c)P(E^c)$$

Since there are four choices, if he guesses randomly, $P(F|E^c) = 0.25$

Hence

$$P(F) = (1)(0.8) + (0.25)(0.2) = 0.85$$

$$P(E|F) = \frac{0.8}{0.85} \approx 0.94$$

That is, if he answers correctly, we are pretty sure that he knows the answer.

The use of Bayes' reasoning can often help us to correctly assess the actual risk, as the next example shows

Example

A laboratory blood test is 95% effective in detecting a certain disease when it is in fact, present. That is, it would give a 5% “false negative” rate. However, the test also yields a “false positive” result for 1% of healthy persons tested. If 0.5% of the population actually has the disease, what is the probability a **person has the disease** given that the **test result is positive**?

Let E be the event that the person has the disease.

Let F be the event that the test result is positive.

$$\begin{aligned} P(E|F) &= \frac{P(EF)}{P(F)} = \frac{P(F|E)P(E)}{P(F|E)P(E) + P(F|E^c)P(E^c)} \\ &= \frac{(0.95)(0.005)}{(0.95)(0.005) + (0.01)(0.995)} \approx 0.323 \end{aligned}$$

which is much lower than expected!

Bayesian Inference

$$P(E|F) = \frac{P(EF)}{P(F)} = \frac{P(EF)}{P(E)} \frac{P(E)}{P(F)} = \frac{P(F|E)P(E)}{P(F)}$$

We often know $P(data|hypothesis)$. However, we would like to know $P(hypothesis|data)$. Bayesian inference enables us to do so by

The diagram illustrates Bayes' Theorem with the following components and arrows:

- Posterior Probability** (red text) with a downward arrow pointing to $P(hypothesis|data)$.
- Likelihood** (red text) with a downward arrow pointing to $P(data|hypothesis)$ in the numerator.
- Prior Probability** (red text) with a downward arrow pointing to $P(hypothesis)$ in the numerator.
- Evidence (also called marginal likelihood)** (red text) with an upward arrow pointing to $P(data)$ in the denominator.

$$P(hypothesis|data) = \frac{P(data|hypothesis)P(hypothesis)}{P(data)}$$

This is the basis of maximum likelihood estimation we shall learn later

If we have a set of hypotheses, and wish to see which hypothesis can best explain the data, we need only to find the hypothesis that maximize $P(hypothesis|data)$. There is no need to compute $P(data)$ since it is constant for every hypothesis

$$P(hypothesis|data) \propto P(data|hypothesis)P(hypothesis)$$

If we have no preference for each hypothesis, then $P(hypothesis)$ is constant

$$P(hypothesis|data) \propto P(data|hypothesis)$$

We find the hypothesis that maximize $P(data|hypothesis)$, i.e., the hypothesis that can best “explain” the data. This is **maximum likelihood estimation** since it maximizes the likelihood function

Example (a more detailed account can be found in <http://jim-stone.staff.shef.ac.uk/BookBayes2012/BayesRuleBookDownloadCh01.html>)

Suppose you wake up one day with spots on your face. Suppose $P(\text{spots}|\text{smallpox}) = 0.9$. Smallpox is a deadly disease, but you are not worried, why?

Smallpox is the first disease to be eradicated from Earth in 1979.

$$P(\text{smallpox}|\text{spots}) = \frac{P(\text{spots}|\text{smallpox})P(\text{smallpox})}{P(\text{spots})}$$

Since $P(\text{smallpox}) \rightarrow 0$, $P(\text{smallpox}|\text{spots}) \rightarrow 0$

General Form of Bayes' Formula

Bayes' formula can be generalized to more than two events. Let $F_1, F_2, \dots, F_n$ be mutually exclusive events such that

$$\bigcup_{i=1}^n F_i = S$$
$$E = \bigcup_{i=1}^n EF_i$$

Since $EF_i, i = 1, \dots, n$ are mutually exclusive,

$$P(E) = \sum_{i=1}^n P(EF_i) = \sum_{i=1}^n P(E|F_i)P(F_i)$$
$$P(F_j|E) = \frac{P(EF_j)}{P(E)} = \frac{P(E|F_j)P(F_j)}{\sum_{i=1}^n P(E|F_i)P(F_i)}$$

Probability in Daily Life



We often intuitively analyze the risks (e.g. COVID 19 scenarios)

Challenging the Intuition

- Some probability questions challenge our intuition. We need to keep a clear head and really apply what we learn to solve the problems we face. Below is one example:

The Monte Hall Problem:

<https://www.youtube.com/watch?v=mhlc7peGlGg>

Other learning material

Bayes' theorem

<https://www.youtube.com/watch?v=HZGCoVF3YvM>